

**Seventh Semester B.E./B.Tech. Degree Examination, Dec.2025/Jan.2026**  
**Big Data Analytics**

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*  
*2. M : Marks , L: Bloom's level , C: Course outcomes.*

| Module – 1 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M  | L  | C   |
|------------|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.1        | a. | Define Big Data Analytics. Explain classification of analytics.                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10 | L2 | CO1 |
|            | b. | List and explain the terminologies used in Big Data Environments.                                                                                                                                                                                                                                                                                                                                                                                                                                  | 10 | L2 | CO1 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |
| Q.2        | a. | Explain Hadoop ecosystem with a neat diagram.                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 10 | L2 | CO1 |
|            | b. | What is NOSQL? Explain the different types of NOSQL databases with an example.                                                                                                                                                                                                                                                                                                                                                                                                                     | 10 | L2 | CO1 |
| Module – 2 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |
| Q.3        | a. | With a neat diagram explain Hadoop distributed file system architecture.                                                                                                                                                                                                                                                                                                                                                                                                                           | 10 | L2 | CO2 |
|            | b. | Give the HDFS commands to perform the following operations:<br>i) To get the list of directories and files at the root of HDFS<br>ii) To create a directory (Say sample) in HDFS<br>iii) To copy a file from HDFS to local file system<br>iv) To display the contents of an HDFS file on console.                                                                                                                                                                                                  | 10 | L3 | CO2 |
| OR         |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |
| Q.4        | a. | Explain anatomy of HDFS file read and write.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 10 | L2 | CO2 |
|            | b. | With a neat diagram, demonstrate YARN architecture.                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10 | L3 | CO2 |
| Module – 3 |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |     |
| Q.5        | a. | What is MongoDB? Explain creation of database, dropping of database and datatypes of MongoDB.                                                                                                                                                                                                                                                                                                                                                                                                      | 10 | L3 | CO3 |
|            | b. | Write the MongoDB commands to perform the following operations:<br>i) To find the number of documents in the students collection.<br>ii) To retrieve the first 3 documents from the students collection where in the grade is VII.<br>iii) To sort the documents from the students collection in the ascending order of StudName.<br>iv) To skip the first 2 documents from the students collection.<br>v) To sort the documents from the students collection in the descending order of StudName. | 10 | L3 | CO3 |

OR

|     |    |                                                                                                                             |    |    |     |
|-----|----|-----------------------------------------------------------------------------------------------------------------------------|----|----|-----|
| Q.6 | a. | Demonstrate the following:<br>i) Aggregate function with an examples.<br>ii) MapReduce function in MongoDB with examples.   | 10 | L3 | CO3 |
|     | b. | Demonstrate the following methods with an example for each:<br>i) Save( )<br>ii) Find( )<br>iii) Update( )<br>iv) Insert( ) | 10 | L3 | CO3 |

## Module – 4

|     |    |                                                              |    |    |     |
|-----|----|--------------------------------------------------------------|----|----|-----|
| Q.7 | a. | What is Hive? Explain Hive architecture with a neat diagram. | 10 | L2 | CO4 |
|     | b. | Explain bucketing with an example.                           | 5  | L2 | CO4 |
|     | c. | What is Pig? Explain the key features of Pig.                | 5  | L2 | CO4 |

OR

|     |    |                                                                        |    |    |     |
|-----|----|------------------------------------------------------------------------|----|----|-----|
| Q.8 | a. | Explain the anatomy of Pig, Pig philosophy and execution modes of Pig. | 10 | L2 | CO4 |
|     | b. | Explain any five relational operators of Pig with an example.          | 10 | L2 | CO4 |

## Module – 5

|     |    |                                                                                 |    |    |     |
|-----|----|---------------------------------------------------------------------------------|----|----|-----|
| Q.9 | a. | Explain the five layer architecture for running applications using spark stack. | 10 | L2 | CO5 |
|     | b. | Explain the main features of spark with a neat diagram.                         | 10 | L2 | CO5 |

OR

|      |    |                                                                          |    |    |     |
|------|----|--------------------------------------------------------------------------|----|----|-----|
| Q.10 | a. | Define text mining. With a neat diagram explain the text mining process. | 10 | L2 | CO5 |
|      | b. | With a neat diagram, explain the three phases of web usage mining.       | 10 | L2 | CO5 |

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